```
clear;
clc;
close all;

% Design Parameters
Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2*pi*fc;

% Analog Low-Pass Filter
b_a = [Omega_c];
a_a = [1 Omega_c];

% ========================================
% 1. IMPULSE INVARIANCE METHOD
% ========================================
[b_imp, a_imp] = impinvar(b_a, a_a, Fs);

% ========================================
% 2. BILINEAR TRANSFORMATION METHOD
% ========================================
% Prewarping for digital cutoff frequency fc
Omega_p = (2/T)*tan(pi*fc/Fs);

b_a_pre = [Omega_p];
a_a_pre = [1 Omega_p];

[b_blt, a_blt] = bilinear(b_a_pre, a_a_pre, Fs);

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================
fprintf('=== EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION
===\n\n');
```

```
=== EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION ===
```

```
fprintf('Sampling Frequency: %.2f Hz\n', Fs);
```

```
Sampling Frequency: 5000.00 Hz
```

```
fprintf('Analog Cutoff Frequency: %.2f rad/s\n', Omega_c);
```

```
Analog Cutoff Frequency: 6283.19 rad/s
```

```
fprintf('Prewarped Analog Frequency: %.2f rad/s\n\n', Omega_p);
```

```
Prewarped Analog Frequency: 7265.43 rad/s
```

```
fprintf('--- IMPULSE INVARIANCE COEFFICIENTS ---\n');
```

```
--- IMPULSE INVARIANCE COEFFICIENTS ---
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_imp);
```

```
    1.2566
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_imp);
```

```
    1.0000   -0.2846
```

```
fprintf('--- BILINEAR TRANSFORMATION COEFFICIENTS ---\n');
```

```
--- BILINEAR TRANSFORMATION COEFFICIENTS ---
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_blt);
```

```
    0.4208    0.4208
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_blt);
```

```
    1.0000   -0.1584
```

```
% ========================================
% FREQUENCY RESPONSE
% ========================================
[H_imp, f] = freqz(b_imp, a_imp, 1024, Fs);
[H_blt, ~] = freqz(b_blt, a_blt, 1024, Fs);

% ========================================
% PLOT MAGNITUDE AND PHASE RESPONSES
% ========================================
figure('Name', 'Impulse Invariance vs Bilinear Transformation');

% Magnitude Response
```

```
subplot(2,1,1);

plot(f, abs(H_imp), 'b', 'LineWidth', 1.5);
hold on;
plot(f, abs(H_blt), 'r--', 'LineWidth', 1.5);
grid on;

title('Magnitude Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Absolute Magnitude');

legend('Impulse Invariance', ...
       'Bilinear Transformation', ...
       'Location', 'best');

xlim([0 Fs/2]);

% Phase Response
subplot(2,1,2);

plot(f, unwrap(angle(H_imp))*180/pi, ...
     'b', 'LineWidth', 1.5);
hold on;

plot(f, unwrap(angle(H_blt))*180/pi, ...
     'r--', 'LineWidth', 1.5);
grid on;

title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

legend('Impulse Invariance', ...
       'Bilinear Transformation', ...
       'Location', 'best');

xlim([0 Fs/2]);

sgtitle('Experiment 3: IIR Filter Design Methods');
```

Experiment 3: IIR Filter Design Methods

